

## 1 Question 1

$w$  in DeepWalk plays a critical role in defining the scope of the "context" used to learn node representations. As formalized in Equation (1), DeepWalk aims to maximize the probability of observing neighbors within a window of size  $w$  around a given vertex  $v_i$  in a random walk. Consequently,  $w$  determines the scale of the structural information captured by the embeddings.

### Small $w$

When  $w$  is set to a small value, the optimization process is restricted to the immediate neighborhood of the vertex  $v_i$ .

- The resulting embeddings primarily capture microscopic structural details. Two nodes will have similar representations only if they are topologically very close.
- This setting may fail to capture broader semantic relationships or global community structures.

### Large $w$

When  $w$  is large, the sliding window includes nodes that may be several hops away from  $v_i$  in the random walk sequence.

- The model captures macroscopic views of the graph. The embeddings tend to reflect higher-order proximity, such as community membership.
- Setting  $w$  too high may introduce noise by forcing the model to consider topologically distant and potentially unrelated nodes as contextually similar.

## 2 Question 2

The two embedding matrices  $X_1$  and  $X_2$  are related by a symmetry.

By observing the columns of the matrices, we can see that the first dimension (column 0) is identical in both matrices, and that the second dimension (column 1) in  $X_2$  is the negation of the second dimension in  $X_1$ . Algebraically,  $X_2$  can be obtained from  $X_1$  via a linear transformation using an orthogonal matrix  $Q$ :

$$X_2 = X_1 Q \quad \text{where} \quad Q = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1)$$

These vectors represent equivalent solutions in the latent space since the dot product, which DeepWalk is based on (through SkipGram), is invariant under orthogonal transformations such as symmetry.

## Question 3

Let  $P$  be a permutation matrix.  $P$  is an orthogonal matrix, meaning  $P^T P = I$  and  $P^{-1} = P^T$ . We want to prove that:

$$GNN(PAP^T, PX) = P \cdot GNN(A, X)$$

Let  $A' = PAP^T$  be the permuted adjacency matrix. Then

$$\tilde{A}' = A' + I = PAP^T + PIP^T = P(A + I)P^T = P\tilde{A}P^T$$

Similarly

$$\tilde{D}' = P\tilde{D}P^T$$

Now, let's compute  $\hat{A}'$ :

$$\begin{aligned} \hat{A}' &= (\tilde{D}')^{-1/2} \tilde{A}' (\tilde{D}')^{-1/2} \\ &= (P\tilde{D}^{-1/2}P^T)(P\tilde{A}P^T)(P\tilde{D}^{-1/2}P^T) \end{aligned}$$

$$\begin{aligned}
&= P\tilde{D}^{-1/2}(P^T P)\tilde{A}(P^T P)\tilde{D}^{-1/2}P^T \\
&= P(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2})P^T \\
&= P\hat{A}P^T
\end{aligned}$$

Now into the GNN function definition:

$$\begin{aligned}
GNN(PAP^T, PX) &= f(\hat{A}'(PX)W) \\
&= f((P\hat{A}P^T)(PX)W)
\end{aligned}$$

By associativity of matrix multiplication and utilizing  $P^T P = I$  again:

$$\begin{aligned}
&= f(P\hat{A}(P^T P)XW) \\
&= f(P(\hat{A}XW))
\end{aligned}$$

Since  $f$  is an element-wise activation function, applying it to permuted rows is equivalent to permuting the rows after applying the function (i.e.,  $f(PM) = Pf(M)$  for any matrix  $M$ ):

$$\begin{aligned}
&= P \cdot f(\hat{A}XW) \\
&= P \cdot GNN(A, X)
\end{aligned}$$

## Question 4

1.

In matrix notation, we can write  $u = \tilde{D}^{1/2}\mathbf{1}$ , where  $\mathbf{1}$  is the vector of all ones. We compute the product  $\hat{A}u$ :

$$\hat{A}u = (\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2})(\tilde{D}^{1/2}\mathbf{1})$$

Since diagonal matrices commute and  $\tilde{D}^{-1/2}\tilde{D}^{1/2} = I$ , this simplifies to:

$$\hat{A}u = \tilde{D}^{-1/2}\tilde{A}\mathbf{1}$$

By definition, the product  $\tilde{A}\mathbf{1}$  sums the rows of the adjacency matrix, which yields the vector of degrees. Thus,  $\tilde{A}\mathbf{1} = \tilde{D}\mathbf{1}$ . Substituting this back:

$$\begin{aligned}
\hat{A}u &= \tilde{D}^{-1/2}(\tilde{D}\mathbf{1}) \\
&= (\tilde{D}^{-1/2}\tilde{D})\mathbf{1} \\
&= \tilde{D}^{1/2}\mathbf{1} \\
&= u
\end{aligned}$$

Since  $\hat{A}u = 1 \cdot u$ ,  $u$  is indeed an eigenvector of  $\hat{A}$  corresponding to the eigenvalue  $\lambda = 1$ .

2.

Since  $\hat{A}$  is a real symmetric matrix, it admits a spectral decomposition of the form:

$$\hat{A} = \sum_{i=1}^n \lambda_i v_i v_i^T \quad (2)$$

where  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$  are the eigenvalues and  $v_i$  are the corresponding orthonormal eigenvectors. We know that:

1.  $\lambda_1 = 1$  (with multiplicity 1).
2. The corresponding eigenvector is proportional to  $u$  (where  $u_i = \sqrt{\tilde{d}_i}$ ). The normalized eigenvector is  $v_1 = \frac{u}{\|u\|_2}$ .

3. For all  $i > 1$ ,  $|\lambda_i| < 1$ .

Now, because  $v_i v_i^T = 0$  if  $i \neq j$  and 1 else:

$$\hat{A}^k = \left( \sum_{i=1}^n \lambda_i v_i v_i^T \right)^k = \sum_{i=1}^n \lambda_i^k v_i v_i^T \quad (3)$$

Taking the limit as  $k \rightarrow \infty$ :

- For  $i = 1$ :  $\lim_{k \rightarrow \infty} \lambda_1^k = 1^k = 1$ .
- For  $i > 1$ : Since  $|\lambda_i| < 1$ , we have  $\lim_{k \rightarrow \infty} \lambda_i^k = 0$ .

Thus,

$$\lim_{k \rightarrow \infty} \hat{A}^k = v_1 v_1^T = \frac{u u^T}{\|u\|_2^2} \quad (4)$$

Finally, substituting this into the expression for the node representations  $Z^{(k)}$ :

$$\begin{aligned} \lim_{k \rightarrow \infty} Z^{(k)} &= \left( \lim_{k \rightarrow \infty} \hat{A}^k \right) XW \\ &= \frac{u u^T}{\|u\|_2^2} XW \\ &= u \left( \frac{u^T XW}{\|u\|_2^2} \right) \end{aligned}$$

Let  $c = \frac{u^T XW}{\|u\|_2^2}$  be a constant row vector. The limit can be written as:

$$Z^{(\infty)} = u \cdot c \quad (5)$$

This means that the representation of each node  $i$  converges to  $u_i c = \sqrt{\tilde{d}_i} c$ , which depends only on the node's degree and not on its initial features or local structure.

### 3

Since the node representations converge to a projection onto  $u$  where  $u_i = \sqrt{\tilde{d}_i}$ , the final embedding only depends on the degree, thus nodes with identical degrees are indistinguishable.